

TechConnect Software Requirements Specification (SRS)

Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents the complete functional and non-functional requirements for **TechConnect** — a digital marketplace that connects skilled freelancers with clients in Cameroon. This document is intended for the development team, quality assurance testers, and project stakeholders to guide implementation and validation of the platform.

1.2 Scope

TechConnect is a web-based freelance marketplace that enables:

- User registration with distinct roles: **Skilled Individual (Freelancer)** and **Client**
- Profile creation with skills, biography, and portfolio elements
- Job posting with budgets, milestones, required skills, and deadlines
- Intelligent job matching and search functionality
- Milestone-based workflow with escrow-protected payments
- 5-star rating and review system for both parties
- Admin dashboard for user management and dispute resolution
- 10% platform commission deducted from each completed transaction

The system operates in **Central African CFA franc (FCFA)** and is accessible via modern web browsers on desktop, tablet, and mobile devices.

1.3 Definitions & Acronyms

Term	Definition
Freelancer	A skilled individual offering services (coder, designer, writer, etc.)
Client	A user who posts jobs and hires freelancers
Admin	Platform manager with moderation privileges
Milestone	A discrete stage of a job with its own deliverable and payment amount
Escrow	A financial arrangement where funds are held by a third party (Nkwa Pay) until conditions are met
Offer	A formal job proposal sent by a client to a freelancer

Term	Definition
Submission	Work deliverables submitted by a freelancer for client review
FCFA	Central African CFA franc (currency)
MVP	Minimum Viable Product

2. Overall Description

2.1 Product Perspective

TechConnect is a **web-based marketplace** implemented as a static HTML/CSS/JavaScript application with client-side storage (localStorage) for the MVP. The system interacts with:

- **Nkwa Pay** (simulated payment gateway) for escrow operations
- **Browser local storage** for data persistence during development

The platform replaces informal, insecure freelance arrangements by providing structured workflows, payment protection, and reputation tracking.

2.2 User Characteristics

User Group	Description	Key Needs
Freelancers	Coders, designers, writers, virtual assistants, data entry specialists	Secure payment, legitimate clients, reputation building, milestone-based work
Clients	Small businesses, startups, NGOs, individual project owners	Reliable workers, transparent progress, pay-only-when-satisfied guarantee
Admin	Platform manager	User moderation, dispute resolution, transaction oversight

All users are assumed to have basic digital literacy, an email address, and internet connectivity.

2.3 Assumptions & Constraints

Assumptions:

- Users possess a valid email address for account registration
- Nkwa Pay API (simulated) is available for payment operations
- Clients are willing to deposit funds before work begins
- All monetary values are in FCFA

Constraints:

- MVP uses client-side storage (no backend database)
- No real payment processing (simulated for demonstration)
- File attachments are stored as base64 (size limitations apply)
- Email verification is simulated (no actual SMTP)

3. Functional Requirements

All functional requirements are expressed using the phrase **"The system shall"**.

3.1 User Registration & Authentication

ID	Requirement
REG-01	The system shall allow a user to register using an email address and password.
REG-02	The system shall require the user to select a role during registration: <code>skilled</code> (freelancer) or <code>client</code> .
REG-03	The system shall validate that the email address is not already registered.
REG-04	The system shall simulate email verification via a one-time code before account activation.
REG-05	The system shall allow a user to log in with email and password.
REG-06	The system shall provide password reset functionality that displays the existing password (simulated).
REG-07	The system shall maintain user session state in local storage.
REG-08	The system shall provide a "Remember me" option for persistent login.

3.2 Profile Management

ID	Requirement
PRO-01	The system shall allow a freelancer to create a public profile containing: full name, biography, profile picture, skills list, education, location, and experience.
PRO-02	The system shall allow a client to create a profile containing: full name, company name, biography, and location.

ID	Requirement
PRO-03	The system shall present a predefined list of skills (e.g., Python, JavaScript, React, UI/UX Design, Data Entry, Typing).
PRO-04	The system shall allow users to add custom skills not present in the predefined list.
PRO-05	The system shall allow users to upload a profile picture (stored as base64).
PRO-06	The system shall allow users to edit and update their profile information at any time.
PRO-07	The system shall display a user's aggregate rating on their profile.
PRO-08	The system shall automatically award a "Top Rated" badge to freelancers with average rating $\geq$ 4.5 stars over at least 10 completed jobs.

3.3 Job Posting

ID	Requirement
JOB-01	The system shall allow a client to post a job with the following mandatory fields: title, description, total budget (FCFA), deadline, required skills (multiple selection), and experience level (beginner/intermediate/advanced).
JOB-02	The system shall validate that the budget is a positive number.
JOB-03	The system shall validate that the deadline is a future date.
JOB-04	The system shall require each job to be divided into one or more milestones.
JOB-05	For each milestone, the system shall require a description and an amount in FCFA.
JOB-06	The system shall validate that the sum of all milestone amounts equals the total job budget.
JOB-07	The system shall store a new job with an initial status of <code>created</code> .
JOB-08	The system shall allow a client to delete their own jobs that are in <code>created</code> or <code>active</code> status.

3.4 Job Discovery & Browsing

ID	Requirement
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ID	Requirement
DISC-01	The system shall display a searchable, filterable list of active jobs to freelancers.
DISC-02	The system shall provide search functionality by job title, description, or required skill keyword.
DISC-03	The system shall provide filtering by required skill and experience level.
DISC-04	The system shall display each job card with: title, budget, truncated description, required skills (up to 3), deadline, and client "Top Rated" badge if applicable.
DISC-05	The system shall display detailed job information on a dedicated job details page, including: full description, all required skills, all milestones, client information, and client rating.

3.5 Hiring & Offers

ID	Requirement
OFF-01	The system shall allow a freelancer to apply for a job by submitting an offer.
OFF-02	The system shall generate an offer containing: job ID, freelancer ID, client ID, proposed price, message, status (pending), creation timestamp, and expiration timestamp (48 hours).
OFF-03	The system shall prevent a freelancer from applying multiple times to the same job.
OFF-04	The system shall allow a client to view all pending offers for their jobs (admin view).
OFF-05	The system shall allow a client to send a formal job offer to a specific freelancer, specifying the job, freelancer email, offer amount, and message.
OFF-06	The system shall allow a freelancer to accept or decline an offer within the 48-hour window.
OFF-07	Upon offer acceptance, the system shall update the job status to accepted and assign the freelancer to the job.
OFF-08	The system shall prevent a freelancer from submitting work for a job unless they have accepted an offer for that job.

3.6 Milestone-Based Workflow

ID	Requirement
MIL-01	The system shall allow a freelancer to submit work deliverables for a specific milestone, including a text description and optional file attachment.

ID	Requirement
MIL-02	The system shall prevent milestone submissions for jobs that are not in accepted status.
MIL-03	The system shall prevent a freelancer from submitting work for a milestone that has already been approved.
MIL-04	The system shall allow a client to review a milestone submission and either "Approve" or "Request Changes".
MIL-05	The system shall require a mandatory comment when requesting changes.
MIL-06	The system shall limit revision rounds to three per milestone.
MIL-07	Upon milestone approval, the system shall record the approval timestamp and trigger payment release (simulated).
MIL-08	Upon approval of the final milestone, the system shall update the job status to completed .
MIL-09	The system shall allow a client to view all pending milestone submissions awaiting their review.

3.7 Payment Integration (Nkwa Pay)

ID	Requirement
PAY-01	The system shall allow a client to choose between "Full Upfront Payment" and "Milestone-Based Payment" when funding a job.
PAY-02	Before work begins, the system shall redirect the client to a payment page where they can deposit the required amount.
PAY-03	The system shall not activate a job (status active) until a successful payment confirmation is simulated.
PAY-04	On successful payment, the system shall: generate a unique transaction ID, store the amount, timestamp, and payment method, and update the job status to active .
PAY-05	The system shall display a payment summary showing: job title, milestone amount (if applicable), platform commission (10%), and total amount to pay.
PAY-06	After a milestone is approved, the system shall simulate fund release to the freelancer.
PAY-07	The system shall automatically deduct a 10% platform commission from the released amount before the freelancer receives funds.
PAY-08	The system shall maintain a log of all payment transactions, including: transaction ID, job ID, amount, commission, total paid, client ID, status, and timestamp.

ID	Requirement
PAY-09	The system shall display a payment confirmation page with transaction details after successful payment.

3.8 Rating & Feedback System

ID	Requirement
RAT-01	The system shall prompt both the client and the freelancer to rate each other after a job is marked <code>completed</code> .
RAT-02	The system shall allow ratings on a 1-5 star scale with an optional written review.
RAT-03	The system shall prevent multiple ratings for the same job by the same user.
RAT-04	The system shall compute an aggregate rating score for each user based on all received ratings.
RAT-05	The system shall display the aggregate rating on user profiles and job cards.
RAT-06	The system shall store each rating with: rating ID, job ID, from user ID, to user ID, score, comment, and timestamp.

3.9 Admin Dashboard

ID	Requirement
ADM-01	The system shall provide an admin dashboard accessible only to users with the <code>admin</code> role.
ADM-02	The admin credentials shall be predefined: <code>Admin01@gmail.com</code> / <code>Admin001</code> .
ADM-03	The system shall allow the admin to view all registered users with their email, role, name, rating, and suspension status.
ADM-04	The system shall allow the admin to suspend or activate any user account.
ADM-05	The system shall allow the admin to delete any user account (cascading to profile, jobs, offers, submissions, and ratings).
ADM-06	The system shall allow the admin to view all jobs with their ID, title, budget, status, disputed flag, client, and assigned freelancer.
ADM-07	The system shall allow the admin to delete any job (cascading to related offers, submissions, and payments).

ID	Requirement
ADM-08	The system shall allow the admin to view all payment transaction logs.
ADM-09	The system shall allow the admin to toggle a "dispute freeze" on any job. When frozen, no milestone approvals or payment releases are permitted.
ADM-10	The system shall allow the admin to clear all application data (factory reset).

3.10 Data Storage Requirements

The system shall persist the following core entities using browser localStorage:

Entity	Key	Attributes
User	<code>techconnect_users</code>	uid, email, password, role, emailVerified, suspended
Profile	<code>techconnect_profiles</code>	uid → name, bio, skills, experience, companyName, clientDesc, profileImageUrl, rating, education, location
Job	<code>techconnect_jobs</code>	id, title, description, budget, deadline, requiredSkills, level, milestones, clientId, skilledUserID, status, disputed, createdAt
Offer	<code>techconnect_offers</code>	id, jobId, jobTitle, clientId, freelancerId, price, message, status, createdAt, expiresAt
Milestone Submission	<code>techconnect_milestone_submissions</code>	id, jobId, milestoneIndex, milestoneDescription, amount, description, status, revisionCount, submittedAt, approvedAt, feedback
Payment	<code>techconnect_payments</code>	id, jobId, milestoneId, amount, commission, totalPaid, clientId, status, paymentMethod, provider, timestamp
Rating	<code>techconnect_ratings</code>	id, jobId, fromUserId, toUserId, score, comment, date
Disbursement	<code>techconnect_disbursements</code>	jobId, milestoneIndex, amount, commission, netAmount, timestamp

4. Non-Functional Requirements

4.1 Security

ID	Requirement
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ID	Requirement
SEC-01	The system shall implement role-based access control for all pages (freelancer, client, admin).
SEC-02	The system shall redirect unauthenticated users to the login page.
SEC-03	The system shall prevent clients from accessing freelancer-only pages and vice versa.
SEC-04	The system shall not expose user passwords in plain text (stored as-is for MVP; production requires hashing).

4.2 Usability & Accessibility

ID	Requirement
UX-01	The interface shall be responsive and usable on screens from 320px (mobile) to 1920px (desktop).
UX-02	The system shall use a consistent color scheme (deep gray, violet, soft white) across all pages.
UX-03	All forms shall provide clear validation error messages.
UX-04	The system shall display notifications for all user actions (success, error, info).
UX-05	The system shall provide a loading indicator during asynchronous operations.
UX-06	Navigation shall be consistent across all pages with a sidebar menu and top navbar.

4.3 Performance

ID	Requirement
PER-01	All web pages shall load within 3 seconds under normal network conditions.
PER-02	The system shall support at least 500 concurrent users without degradation (theoretical for MVP).
PER-03	Search and filter operations shall respond within 1 second.

4.4 Reliability

ID	Requirement
REL-01	The system shall maintain data integrity across localStorage operations.
REL-02	The system shall validate all user inputs before processing.
REL-03	The system shall handle missing or corrupted data gracefully.
REL-04	The system shall provide visual feedback for all user actions.

4.5 Maintainability

ID	Requirement
MAI-01	The codebase shall be organized by feature modules (auth, profile, jobs, offers, payments, admin).
MAI-02	Styling shall be centralized in a common CSS file (<code>common.css</code>).
MAI-03	The system shall use consistent naming conventions for localStorage keys.

5. User Interface Requirements

5.1 Page Structure

The system shall consist of the following pages:

Page	File	Access
Landing page	<code>index.html</code>	Public
Registration	<code>Feature 1 - Register.html</code>	Public
Login	<code>Feature 2 - Login.html</code>	Public
Password reset	<code>Feature 4 - Reset Password.html</code>	Public
Profile	<code>Feature 3 - Profile.html</code>	Authenticated
Home dashboard	<code>home.html</code>	Authenticated
Browse jobs	<code>dashboard.html</code>	Freelancer
Job details	<code>job-details.html</code>	Authenticated
Post job	<code>post-job.html</code>	Client
Send offer	<code>send-offer.html</code>	Client

Page	File	Access
Pending offers	my-jobs.html	Freelancer
Submit milestone	submit-milestone.html	Freelancer
Review milestone	review-milestone.html	Client
Make payment	payment.html	Client
Payment confirmation	payment-confirmation.html	Client
Rate & feedback	feedback.html	Authenticated
Admin dashboard	admin.html	Admin only
FAQ	faq.html	Public

5.2 Visual Design System

Element	Specification
Primary color	#8E44AD (violet)
Secondary color	#1E1E1E (deep gray)
Background	#EAEAEA (soft white)
Card background	#FFFFFF
Border radius	8px-28px (component-dependent)
Typography headings	'Syne', sans-serif
Typography body	'Space Grotesk', sans-serif
Buttons	Pill-shaped with hover effects

6. System Requirements

6.1 Client-Side Requirements

Component	Requirement
Browser	Modern browser with JavaScript enabled (Chrome, Firefox, Edge, Safari latest versions)
Device	Desktop, tablet, or smartphone with minimum 320px screen width
RAM	Minimum 4 GB recommended

Component	Requirement
Internet	Stable connection (broadband or 3G mobile)

6.2 Development Environment

Tool	Purpose
VS Code / any editor	Code development
Chrome DevTools	Testing and debugging
Git	Version control (optional)
LocalStorage API	Data persistence

7. Future Enhancements

The following features are planned for future releases:

Feature	Description
Real-time chat	Direct messaging between clients and freelancers
Mobile apps	Native Android and iOS applications
Multiple payment gateways	MTN Mobile Money, Orange Money integration
Identity verification	Government ID upload with "Verified" badge
Skill assessments	Certified badges for high-demand skills
Tiered commission	Reduced commission rates for high-volume users
Email notifications	Real email notifications via SMTP
Backend database	Migration from localStorage to cloud database
Advanced matching	AI-powered job recommendations

Appendix A: Sample Data

The system shall include the following sample jobs for demonstration purposes:

Title	Budget	Skills	Level
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Title	Budget	Skills	Level
Data Entry Clerk	25,000 FCFA	Typing, Data Entry, Excel	Beginner
Typing Job - Document Transcription	15,000 FCFA	Typing, Word	Beginner
E-commerce Website Development	350,000 FCFA	JavaScript, React, Node.js	Advanced
UI/UX Design for SaaS Platform	250,000 FCFA	UI/UX Design, Figma	Intermediate

Appendix B: Revision History

Version	Date	Author	Description
1.0	2026	TechConnect Development Team	Initial release based on implemented MVP

Prepared for: TechConnect Development Team
Based on: Implemented web application (HTML/CSS/JS with localStorage)
Document Status: Final